FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CONCEPT RECAPITULATION TEST – II
PAPER-2

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**D**	24.	**C**	47.	**B**
2.	**D**	25.	**B**	48.	**A**
3.	**A**	26.	**B**	49.	**A**
4.	**BD**	27.	**ABC**	50.	**CD**
5.	**ACD**	28.	**AB**	51.	**AB**
6.	**ABCD**	29.	**ABC**	52.	**ABD**
7.	**ACD**	30.	**AB**	53.	**ABC**
8.	**ACD**	31.	**BD**	54.	**ABCD**
9.	**A**	32.	**C**	55.	**A**
10.	**D**	33.	**D**	56.	**C**
11.	**1**	34.	**7**	57.	**1**
12.	**1**	35.	**4**	58.	**2**
13.	**1**	36.	**3**	59.	**5**
14.	**6**	37.	**3**	60.	**3**
15.	**4**	38.	**4**	61.	**3**
16.	**7**	39.	**3**	62.	**5**
17.	**6**	40.	**4**	63.	**4**
18.	**2**	41.	**6**	64.	**7**
19.	**6**	42.	**7**	65.	**3**
20.	**4**	43.	**3**	66.	**1**
21.	**00003.73**	44.	**00015.20**	67.	**24850.00**
22.	**00629.81**	45.	**00012.60**	68.	**26916.72**
23.	**00629.20**	46.	**00116.24**	69.	**24869.83**

Physics

PART – I

SECTION – A

1. D

Sol. Impulse on ball

$I = (1+0.5)\sqrt{2gh}\ m$ (vertically)

Frictional impulse on block (horizontally)

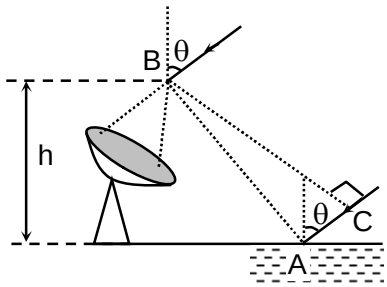
$= \mu I = 0.2 \times 1.5\ m\sqrt{2gh}$

Decrease in velocity

$\Delta V = \frac{\mu I}{m} = 0.3\sqrt{2gh}$

2. D

Sol. Path difference of the two rays (one coming directly to antenna and other after reflection from water surface



$$\Delta x = AB + AC + \frac{\lambda}{2}$$

$$\Rightarrow \quad \Delta x = \frac{h}{\cos\theta} + \frac{h}{\cos\theta}\cos 2\theta + \frac{\lambda}{2}$$

$$= \quad \frac{h}{\cos\theta}(1+\cos 2\theta) + \frac{\lambda}{2}$$

or $\Delta x = 2h\cos\theta + \frac{\lambda}{2}$

For constructive interference $\Delta x = n\lambda$

$$\Rightarrow \quad 2h\cos\theta + \frac{\lambda}{2} = n\lambda \quad \text{or} \quad h = \frac{\left(n-\frac{1}{2}\right)\lambda}{2\cos\theta} = \frac{\lambda}{4\cos\theta}, \frac{3\lambda}{4\cos\theta}, \ldots\ldots \text{ etc}$$

3. A

Sol. From first law of thermodynamics

$\Delta Q = \Delta U + \Delta W$

$$\Delta Q = nC_V(T_2 - T_1) + \frac{nR(T_1 - T_2)}{x-1}$$

$$\frac{1}{n}\left(\frac{\Delta Q}{\Delta T}\right) = C_V - \frac{R}{x-1}$$

$\therefore \quad (C - C_V)(x-1) = -R$

$\therefore$ graph is a rectangular, hyperbola

$\therefore$ co–ordinate of P_1 (O, C_P)

i.e. $\left(0, \frac{5}{2}R\right)$ and that of $P_4\left(\frac{5}{3}, 0\right)$

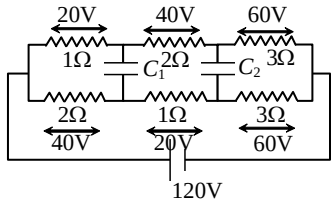
4. BD

Sol. For n^{th} mode of vibration $\lambda_n = \dfrac{4\ell}{(2n-1)}$.

5. ACD

Sol. Now, potential difference across C_1 is 20 V and across C_2 is zero.

$\therefore$ charge stored in C_1 is 40 μC and in C_2 is zero.



6. ABCD

Sol. $x + \alpha = \beta + x = 90^0 \Rightarrow \alpha = \beta$

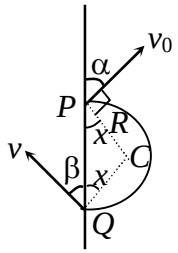
Also, $PQ = 2R\cos(90-\alpha) = 2R\sin\alpha = \dfrac{2mv_0}{qB}\sin\alpha$

$|\vec{v}| = \text{constant}$

$\therefore \quad v = v_0$

For 2π rotation it takes $\dfrac{2\pi m}{qB}$ time

For $(2\pi - 2\alpha)$ it takes $\dfrac{2m(\pi-\alpha)}{qB}$



7. ACD

Sol. The gravitational force due to these masses on a mass at *O* is equal and opposite.
So, the resultant force is zero hence, the resultant field is zero
Also, any point on *y-z* plane which is equidistant from two cavities will have zero field intensity hence constant potential.

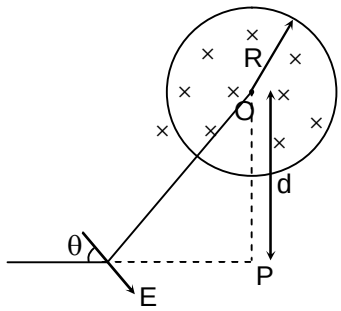
8. ACD

Sol. $\int \vec{E}\cdot d\vec{l} = A\dfrac{dB}{dt}$

$E2\pi\sqrt{x^2+d^2} = \pi R^2 k$

$E = \dfrac{\pi R^2 k}{2\sqrt{x^2+d^2}}$

$W_{ext} = \int_0^\infty q\vec{E}\cdot dx = \dfrac{q\pi R^2}{4}k$



9. A

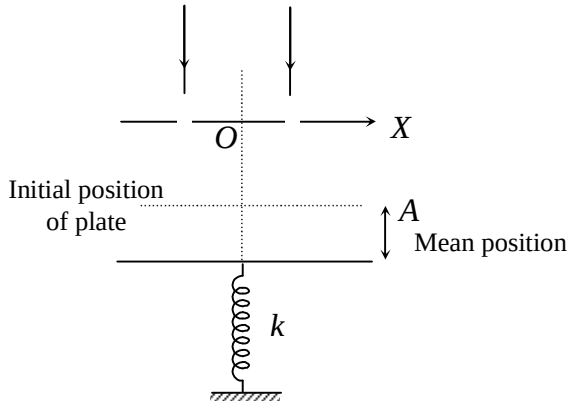
Sol. $\beta = \dfrac{(D+y)\lambda}{d}$

$\dfrac{d\beta}{dt} = \dfrac{\lambda}{d}\dfrac{dy}{dt} = \dfrac{\lambda}{d}v$

At mean position

$v = A\omega = \dfrac{mg}{k}\sqrt{\dfrac{k}{m}} = g\sqrt{\dfrac{m}{k}}$

$\dfrac{d\beta}{dt} = \dfrac{\lambda g}{d}\sqrt{\dfrac{m}{k}}$



10. D

Sol. Shift $\dfrac{D(\mu-1)t}{d} = \dfrac{D\lambda}{d}$

$t = \dfrac{\lambda}{\mu-1}$

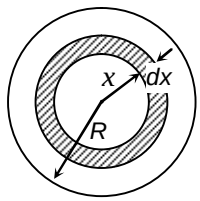
SECTION – C

11. 1

Sol. $v_0 \le \dfrac{gR}{3}(7\cos\alpha - 4)$

12. 1

Sol. $d\phi = (2\pi x dx)kxt^2$



$\phi = \dfrac{2}{3}kt^2\pi x^3$

$\dfrac{d\phi}{dt} = \dfrac{4k\pi t x^3}{3}$

$E2\pi x = \dfrac{4\pi k t x^3}{3}$; $E = \dfrac{2}{3}ktx^2$; $d\tau = \left(\dfrac{2}{3}ktx^2\right)\dfrac{Q}{\pi R^2}(2\pi x dx)x$

$\int d\tau = \dfrac{4}{3}\dfrac{ktQ}{R^2}\int_0^R x^4 dx$ $\Rightarrow \tau = \dfrac{4}{3}\dfrac{ktQ}{R^2}\dfrac{R^5}{5}$

At t = 15sec, τ = 1 N-m

13. 1

Sol. $a = \omega^2 A$

$f_1 = (1)\omega^2 A \le 6$ …(1)

$f_2 = 3\omega^2 A \le 12$

$\omega^2 A \le 4$ …(2)

$\Rightarrow$ $\omega^2 A = 4$

$$A = \frac{4}{\left(\frac{k}{m}\right)} = \frac{4\times 6}{24} = 1$$

14. 6

Sol. Energy in $2n^{th}$ state will be $= -\frac{13.6Z^2}{4n^2}eV$

Max energy of photon = $13.6Z^2\left(1-\frac{1}{4n^2}\right) = 204$ ....(1)

Energy of photon when it makes transition to x in state will be

$$13.6\,Z^2\left(\frac{1}{n2}-\frac{1}{4n2}\right) = 40.8$$

.....(2)

Dividing equation (1) and (2)

$$\frac{4n^2-1}{3} = 5$$

$$\Rightarrow \quad n^2 = 4 \; n = 2$$

Putting of $n = 2$ in equation (2)

$$13.6\,Z^2\left(\frac{3}{4\times 4}\right) = 40.8$$

$$Z^2 = \frac{40.8\times 16}{3\times 13.6}$$

$$Z = 4$$

15. 4

Sol. Since rate of heat flow remains same in both the cases, so

$$\int_R^{2R}\frac{dx}{k2\pi x l} = \int_{\frac{R}{4}}^{R}\frac{dx}{nk\left(2\pi x\right)\frac{l}{2}} \quad \Rightarrow \quad nk = 4k$$

$$\Rightarrow \; n = 4.$$

16. 7

Sol. For process AB $\quad T_A = 300\ K\,,\ T_B = 600\ K$

$$W = nR\Delta T = nR(T_B - T_A) = 300\ nR\ = 600R.$$

$$Q = n\ Cp\,\Delta T = 2\times\frac{5}{2}R\ (300) = 1500R.$$

For process BC $\quad W = nRT\ln\frac{v_f}{v_c} = nRT\ln\frac{p_i}{p_f}\ = nRT\ln 2\ = 1200R\ln 2$

$$Q = W = 1200R\ln 2$$

For process CA $\quad W = \int P\ dV = \int_{600}^{300}\frac{K}{T}\frac{2nRT}{K}dT.$

$$= -2nR(300) = -1200R.$$

$$Q = nC_V\ \Delta\ T + W$$

$$= 2\times\frac{3}{2}R(-300) - 1200R.$$

$$= -900R - 1200R = -2100R$$

$$\eta = \frac{600R + 1200R\ln 2 - 1200R}{1500R + 1200R\ln 2}$$

$$= 1 - \frac{21}{12\ln 2 + 15} \quad \Rightarrow \ x = 7.$$

17. 6

Sol. As source (horn of bus) is approaching stationary wall (say, listener), therefore, apparent frequency striking the wall is

$$v' = \frac{\upsilon\, v}{\upsilon - \upsilon_s} \qquad \ldots(1)$$

Sound of this frequency will be reflected by the wall (now, source). The passenger is the listener moving towards source. Therefore, frequency heard by the listener

$$v'' = \frac{(\upsilon + \upsilon_L)v'}{\upsilon}$$

using (1)

$$v'' = \frac{\upsilon + \upsilon_L}{\upsilon} \times \frac{\upsilon v}{\upsilon - \upsilon_s} = \frac{(\upsilon + \upsilon_L)v}{\upsilon - \upsilon_s}$$

$$= \frac{(330 + 5) \times 200}{330 - 5} = \frac{335}{325} \times 200$$

$$v'' = 206\,\text{Hz}$$

$\therefore$ Beat frequency = $(v'' - v)$

$$= 206 - 200 = 6\,\text{Hz}$$

18. 2

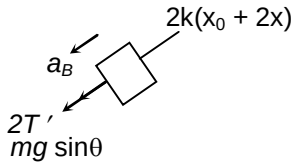
Sol. Let elongation of spring be x_0 in equilibrium. Then,

$$2T + mg\sin\theta = 2kx_0 \qquad \ldots(i)$$

and $$T = mg \qquad \ldots(ii)$$

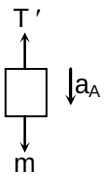
Let Block B is displaced by x down the inclination

F.B.D. of B



$$-ma_B = 2k(x_0 + 2x) - 2T' - mg\sin\theta \qquad \ldots(iii)$$

F.B.D. of *A*



$mg - T' = ma_A$

Also, $a_A = 2a_B$

$$T' = mg - 2ma_B$$

$$-ma_B = 2kx_0 + 4kx - 2mg + 4ma_B - mg\sin\theta$$

$$-ma_B = 4kx + 4ma_B$$

$$a_B = -\frac{4k}{5m}x$$

$$\therefore\ T = 2\pi\sqrt{\frac{5m}{4k}}$$

T = 6.28 s.

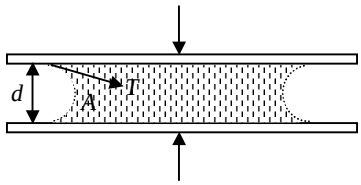
19. 6

Sol. Let R be the radius of the circular layer of water.

Then, $\pi R^2 d \times \rho = m$ …(1)

Pressure at $A = p_0 - \frac{2T}{d}$ …(2)

Thus pressure between the plates is less than the atmospheric pressure and so the plates are pressed together as though attracted towards each other.



F, force of attraction = $\Delta p \times$ area $\Rightarrow\ F = \frac{2T}{d} \times \pi R^2$

$$\Rightarrow \qquad F = \frac{2T}{d} \times \frac{m}{d\cdot\rho} = \frac{2Tm}{d^2\cdot\rho} = \frac{2\times 0.4\times 10^{-3}\times 0.08}{0.01^2\times 10^{-4}\times 1000} = 6.4\ \text{N}$$

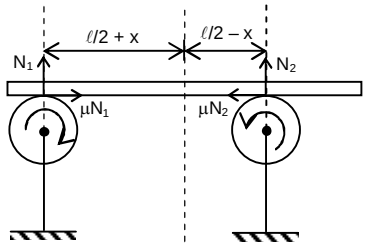
20. 4

Sol. $N_1 + N_2 = mg$

$N_1\ell = mg(\ell/2 - x)$

$$N_1 = \frac{mg}{\ell}\left(\ell/2 - x\right)$$

$$N_2 = \frac{mg}{\ell}\left(\ell/2 + x\right)$$



$$F_{res} = \mu N_2 - \mu N_1 = \mu\left(\frac{2mgx}{\ell}\right)$$

$$\Rightarrow\ \omega = \sqrt{\frac{2\mu g}{\ell}}\ ;\quad T = 2\pi\sqrt{\frac{\ell}{2\mu g}}$$

SECTION – D

21. 00003.73

Sol. $V\left(x = \frac{A}{2}\right) = \sqrt{3}\,\omega\frac{A}{2}$

$$T_1 = 2\times\left(\frac{T}{4} + \frac{T}{12}\right) = \frac{2T}{3} \qquad \left(\text{SHM Part} : T = \frac{2\pi}{\omega}\right)$$

$$T_2 = 2\times\frac{\sqrt{3}\,\omega\frac{A}{2}}{\frac{A\omega^2}{2}} = \frac{2\sqrt{3}}{\omega} \qquad \text{(Constant acceleration part)}$$

Time period $= T_1 + T_2$

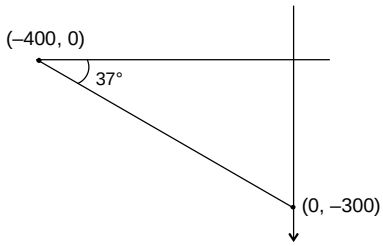
$$= \frac{4\pi}{300} + \frac{2\sqrt{3}}{\omega}$$

22. 00629.81

Sol. $V_{source} = 10\,\hat{i}$, $V_{observer} = -10\,\hat{i}$

$$f_{app} = 600\ Hz \times \frac{330 + 10.cs\ 37^o}{330 - 10.\cos 37^o}$$

$= 629.81$ Hz



23. 00629.20

Sol. $X_L = 50\ \Omega$, $X_C = 100\ \Omega$

$Z_1 = 100 - 50\ I$; $Z_2 = 100 + 100\ i$ $(i = \sqrt{-1})$

$$Z = \frac{Z_1 Z_2}{Z_1 + Z_2} = \frac{100}{17}(13 + i)$$

$$|Z| = \frac{100}{17}\sqrt{170}\ ;\ \cos\theta = \frac{13}{\sqrt{170}}$$

$$P = \frac{V^2}{|Z|}.\cos\phi = 629.20$$

Chemistry

PART – II

SECTION – A

24. C
Sol. The orbitals are 4s(one), 4p(three) and 4d(three). Total no. of electrons = $2 \times 7 = 14$.

25. B
Sol. It is a first order reaction.

26. B
Sol. Both NO and O_2 are formed at the same time.

27. ABC
Sol. The balanced equation is
$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \longrightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$

28. AB
Sol Weak acids form buffer solutions with strong bases.

29. ABC
Sol. O and O_2 contain two unpaired electrons each. NO contains one unpaired electron. N_2 contains no unpaired electrons.

30. AB
Sol. Bases can form silicates with SiO_2(an acidic compound).

31. BD
Sol. $(BN)_x$ and H_3BO_3 form polymers.

32. C
33. D
(Sol. for 32 & 33)
It has maximum number of α-hydrogen atoms which display hyperconjugation.

SECTION – C

34. 7
Sol. $X = (NH_4)_2Cr_2O_7$, $Y = Cr_2O_3$, $Z = N_2$, the liquid is H_2O.

35. 4
Sol. $\Delta T_b = K_b \times i \times m$
or, $4.16 = 0.52 \times i \times 2$
or $i = 4$

36. 3
Sol. $\Delta H = nC_v \Delta T = 1 \times \frac{3}{2} R \times (1300 - 300) = \frac{3}{2} \times 2 \times 1000 = 3000$ cal $= 3$ Kcal.

37. 3

Sol. $E_{Cell} = E^{o}_{Cell} - \frac{0.0591}{n}\log\frac{\left[Zn^{2+}\right]}{\left[H^{+}\right]^{2}}$

or, $0.6418 = [0-(-0.76)] - \frac{0.0591}{n}\log\frac{10^{-2}}{\left[H^{+}\right]^{2}}$

on solving, $[H^+]^2 = 10^{-6}$

$\therefore [H^+] = 10^{-3}$, pH = 3

38. 4

Sol. n-factor of NH_4NO_3 is 4.

39. 3

Sol. $\frac{r_{SO_2}}{r_{CH_4}} = \sqrt{\frac{M_{CH_4}}{M_{SO_2}}} = \sqrt{\frac{16}{64}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$

$\therefore$ x : y = 1 : 2, x + y = 3

40. 4

Sol. FCC contains four octahedral voids and eight tetrahedral voids.

41. 6

Sol. $Cr_2(SO_4)_3 \xrightarrow{NaOH} Cr(OH)_3 \xrightarrow{Na_2O_2} Na_2CrO_4$

42. 7

Sol. No. of ionization isomers = 3

No. of linkage isomers = 4

43. 3

Sol. The products those give Tollen's test are HCHO, HCOOH and CH_3CHO.

SECTION – D

44. 00015.20

Sol. $X(g) \rightleftharpoons Y(g) + Z(g)$

Initial conc. $\frac{250}{100} = 2.5$

At equ^m 2.5 – 1.25 1.25 1.25

= 1.25

$X(g) \rightleftharpoons Y(g) + Z(g)$

At equ^m 1.25 1.25 1.25

After addition (1.25 + x - y) (1.25 + y) (1.25 + y)

1.25 + y = 1.3

or, y = 0.05

$\therefore K_C = \frac{1.3 \times 1.3}{1.25 + x - 0.05} = 1.25$

On solving, x = 0.152

$$0.152 = \frac{n}{V} = \frac{n}{100\,L}$$

or, n = 15.20

45. 00012.60

Sol. $pH = \frac{1}{2}\left[p_{K_w} + p_{k_a} + \log C\right]$

$$\text{or, } 12.8 = \frac{1}{2}\left[14 + p_{k_a} + \log 10^{-1}\right]$$

On solving, $p^{K_a} = 12.60$

46. 00116.24

Sol. $\Delta T_f = K_f \times m$

$$\text{or, } 0 - T_f = 1.86 \times 2 \times \frac{1000}{W}$$

$$\text{or, } 0 - (-4.8) = 1.86 \times \frac{2000}{W}$$

On solving, W = 775 g

$\therefore$ Mass of ice formed = 891.24 – 775 = 116.24 g

Mathematics

PART – III

SECTION – A

47. B

Sol. Writing r as linear combination of a, b and $a \times b$, we have $r = xa + yb + z(a \times b)$ for solats x, y, z

$0 = r.a = x|a|^2 + ya.b$

$1 = r.b = xa.b + y|b|^2$

Solving we get $y = \dfrac{|a|^2}{|a|^2|b|^2 - (a.b)^2} = |a|^2$ and $x = \dfrac{a.b}{(a.b)^2 - |a|^2|b|^2} = a.b$

Also $1 = [r \quad a \quad b] = z|a \times b|^2$

$\Rightarrow \qquad z = \dfrac{1}{|a \times b|^2}$

Thus $r = \left((a.b)a - |a|^2 b\right) + \dfrac{a \times b}{|a \times b|^2}$

$= a \times (a \times b) + \dfrac{a \times b}{|a \times b|^2}$

48. A

Sol. Integrating the given differential equation, we have $\dfrac{dy}{dx} = \dfrac{-\cos 3x}{3} + e^x + \dfrac{x^3}{3} + C_1$ but

$y_1(0) = 1$ so $1 = \left(\dfrac{-1}{3}\right) + 1 + C_1 \Rightarrow C_1 = \dfrac{1}{3}$.

Again integrating, we get $y = \dfrac{-\sin 3x}{9} + e^x + \dfrac{x^4}{12} + \dfrac{1}{3}x + C_2$

but $y(0) = 0$ so $0 = 0 + 1 + C_2 \Rightarrow C_2 = -1$. Thus $y = \dfrac{-\sin 3x}{9} + e^x + \dfrac{x^4}{12} + \dfrac{1}{3}x - 1$

49. A

Sol. $x \neq (2n+1)\dfrac{\pi}{2}$, $n\pi$, $n \in I$. The given inequality can be written as

$$\frac{\log_2\left(x^2 - 8x + 23\right)}{\log_2|\sin x|} > \frac{3}{\log_2|\sin x|}$$

As $\log_2|\sin x| < 0$, we get $\log_2\left(x^2 - 8x + 23\right) < 3$

$\Rightarrow \qquad x^2 - 8x + 23 < 2^3 = 8$

$\Rightarrow \qquad x^2 - 8x + 15 < 0$

$\Rightarrow \qquad (x-5)(x-3) < 0 \Rightarrow 3 < x < 5$

For $x \in (3, 5), x \neq \pi, \frac{\pi}{2}, \frac{3\pi}{2}$. Hence

$x \in (3, \pi) \cup \left(\pi, \frac{3\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 5\right)$

50. CD

Sol. Equating coefficients of x^{20}, we get $2^{20} - a^{20} = 1$

$\Rightarrow \quad a = \pm\left(2^{20} - 1\right)^{1/20}$

Next, putting $x = \frac{1}{2}$ we get $\left(\frac{a}{2} + b\right)^2 + \left(\frac{1}{4} + \frac{1}{2}p + q\right)^{10} = 0$

$\Rightarrow \quad b = \frac{-a}{2}$

The given equation can now be written as $2^{20}\left(x - \frac{1}{2}\right)^{20} - a^{20}\left(x - \frac{1}{2}\right)^{20}$

$= \left(x^2 + px + q\right)^{10}$

or $\left(x - \frac{1}{2}\right)^2 = x^2 + px + q$

$\Rightarrow \quad p = -1, q = \frac{1}{4}$

51. AB

Sol. Note that $S \geq 0$ and $S = 0$ for $z_1 = z_2 = z_3$

Thus, $\alpha = 0$

Since $|a - b|^2 = |a|^2 + |b|^2 - \bar{a}b - a\bar{b}$, we get

$|z_2 - z_3|^2 + |z_3 - z_1|^2 + |z_1 - z_2|^2$

$= 6 - \left(z_2\bar{z}_3 + \bar{z}_2 z_3 + z_3\bar{z}_1 + \bar{z}_3 z_1 + z_1\bar{z}_2 + \bar{z}_1 z_2\right)$

$= 6 - \bar{z}_1(z_2 + z_3) - \bar{z}_2(z_3 + z_1) - \bar{z}_3(z_1 + z_2)$ (1)

Also $0 \leq |z_1 + z_2 + z_3|^2$

$= |z_1|^2 + |z_2|^2 + |z_3|^2 \overline{z_1}(z_2 + z_3) + \bar{z}_2(z_3 + z_1) + \bar{z}_3(z_1 + z_3)$

$\Rightarrow \overline{z_1}(z_2 + z_3) + \overline{z_2}(z_3 + z_1) + \overline{z_3}(z_1 + z_2) \geq -3$ (2)

Thus, from (1) and (2), we get

$|z_2 - z_3|^2 + |z_3 - z_1|^2 + |z_1 - z_2|^2 \leq 9$

The value 9 is attained when $z_1 = 1 \quad z_2 = \omega \quad z_3 = \omega^2$

where $\omega \neq 1$ is a cube root of unity.

52. ABD

Sol. The total number of ways of choosing the ticket is 1000. Let the four digits number on the ticket is 1000.

Let the four digits number on the ticket be $x_1x_2x_3x_4$.

Note that $0 \le x_1 + x_2 \le 18$ and $0 \le x_3 + x_4 \le 18$.

Also, the number of non – negative integral solutions of $x + y = m$ (with $0 \le x, y \le 9$) is $m+1$ if $0 \le m \le 9$ and is 19 – m if $10 \le m \le 18$.

Thus, the number of favourable ways

$= 1\times1\times1 + 2\times2 + ... + 10\times10 + 9\times9 + 8\times8 + ... + 1\times1$

$= 2\left\{\frac{9\times10\times19}{6}\right\} + 100 = 670$

$\therefore$ Probability of required event $= \frac{670}{1000} = 0.067$

Note that $0.067 > 0.065$ and $0.067 < 0.068$.

53. ABC

Sol. $\frac{1}{n+1} + \frac{1}{n+2} + ... + \frac{1}{2n}$

$= \left(1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{2n}\right) - \left(1 + \frac{1}{2} + ... + \frac{1}{n}\right)$

$= \left(1 + \frac{1}{3} + ... + \frac{1}{2n-1}\right)$

$+ \frac{1}{2}\left(1 + \frac{1}{2} + ... + \frac{1}{n}\right) - \left(1 + \frac{1}{2} + ... + \frac{1}{n}\right)$

$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} - ..... + \frac{1}{2n-1} - \frac{1}{2n}$

$\alpha(2n) < \underbrace{\frac{1}{n} + \frac{1}{n} + ... + \frac{1}{n}}_{\text{n times}} = 1$ and $\alpha(2n) \ge \frac{1}{2n} \ge \frac{1}{2} \forall n$

(D) is not correct as $\alpha(1) = 1$

54. ABCD

Sol. Putting $x = 0, \frac{1}{2},$ we obtain $|c| \le 1, |a+b+c| \le 1, \left|\frac{1}{4}a + \frac{1}{2}b + c\right| \le 1$

$\Rightarrow \quad -1 \le c \le 1, -1 \le a+b+c \le 1$

$-4 \le a + 2b + 4c \le 4$

$\Rightarrow \quad -4 \le 4a + 4b + 4c \le 4$

and $\quad -4 \le -a - 2b - 4c \le 4$

Adding we get $-8 \le 3a + 2b \le 8$

Also $\quad -8 \le a + 2b \le 8$

$\therefore \quad -16 \le 2a \le 16 \Rightarrow |a| \le 8$

Since $\quad -1 \le -c \le 1, -8 \le -a \le 8,$

We get $\quad -16 \le 2b \le 16 \Rightarrow |b| \le 8$

Thus, $\quad |a| + |b| + |c| \le 17$

55. A
56. C

Sol. (Q. 55 – 56)

$0 < \alpha < 90^o,\ 90^o < \beta < 180^o,\ 180^o < \gamma < 270^o,\ 270^o < \delta < 360^o$

$\Rightarrow 270^o < \alpha + \delta < 450^o \Rightarrow \alpha + \delta$ lies in the I or IV quadrant and cosine in both is positive.

If d is the common ratio of the A.P.

Then $\beta = \alpha + d,\ \gamma = \alpha + 2d,\ \delta = \alpha + 3d$.

$\Rightarrow \beta + \gamma = \alpha + \delta,\ 2(\alpha - \beta) = -2d = \beta - \delta$ and $\alpha + \gamma = 2\beta$.

Note $\beta - \gamma = -d,\ \alpha - \delta = -3d$

$270^o < \delta < 360^o$

$\Rightarrow \quad 270^o < \alpha + 3d < 360^o$

$\Rightarrow \quad 200^o < 3d < 290^o \qquad [\because \alpha = 70^o]$

$\Rightarrow \quad 400^o < 6d < 580^o$

$\Rightarrow \quad 680^o < 4\alpha + 6d < 860^o$

$\Rightarrow \quad 680^o < \theta < 860^o$

SECTION – C

57. 1

Sol. Applying $C_1 \to C_2 - 2\sin x\, C_3$ and $C_2 \to C_2 + 2\cos x\, C_3$, we obtain

$$f(x) = \begin{vmatrix} 2 & 0 & -\sin x \\ 0 & 2 & \cos x \\ \sin x & -\cos x & 0 \end{vmatrix}$$

$= 2\sin^2 x + 2\cos^2 x = 2 \Rightarrow f'(x) = 0$

$\therefore \int_0^{\pi/2} \left[f(x) + f'(x)\right] dx = \int_0^{\pi/2} 2\,dx = 2\left(\frac{\pi}{2}\right) = \pi$

58. 2

Sol. As $A^2 = O,\ A^k = O\ \forall k \geq 2$

Thus, $(A + I)^{50} = I + 50A$

$\Rightarrow (A + I)^{50} - 50A = I$

$\therefore\ a = 1,\ b = 0,\ c = 0,\ d = 1$

59. 5

Sol. $f(n) = \sum_{j=1}^{n} \left({}^nC_j\right)\left({}^jC_1\right) + \sum_{j=2}^{n} \left({}^nC_j\right)\left({}^jC_2\right) + ... + \sum_{j=n-1}^{n} \left({}^nC_j\right)\left({}^jC_{n-1}\right) + \left({}^nC_n\right)\left({}^nC_n\right)$

$= \left({}^nC_1\right)\left({}^1C_1\right) + \left({}^nC_2\right)\left({}^2C_1 + {}^2C_2\right) + \left({}^nC_3\right)\left({}^3C_1 + {}^3C_2 + {}^3C_3\right)$

$+ ..... + \left({}^nC_n\right)\left({}^nC_1 + {}^nC_2 + ... + {}^nC_n\right)$

$= \left({}^nC_1\right)(2-1) + \left({}^nC_2\right)\left(2^2 - 1\right) + \left({}^nC_3\right)\left(2^3 - 1\right) + ... + {}^nC_n\left(2^n - 1\right)$

$= \left\{(2+1)^n - 1\right\} + \left\{(1+1)^n - 1\right\} = 3^n - 2^n$

60. 3

Sol. Equation of the chord of contact of a point $P(3\sec\theta, 2\tan\theta)$ on the hyperbola with respect to the circle is $(3\sec\theta)x + (2\tan\theta)y = 9$ (1)

Let M (h, k) be the mid point of (1), then equation of (1) in terms of the mid – point is $hx + ky = h^2 + k^2$ (2)

Since (1) and (2) represent the same line.

$\sec\theta = \dfrac{3h}{h^2+k^2}$, $\tan\theta = \dfrac{9k}{2\left(h^2+k^2\right)}$ s

$\Rightarrow$ locus of (h, k) is

$\Rightarrow \dfrac{9x^2}{\left(x^2+y^2\right)^2} - \dfrac{81y^2}{4\left(x^2+y^2\right)^2} = 1$

or $4\left(x^2+y^2\right)^2 = bx^2 - cy^2$

$\Rightarrow$ $a = 4, b = 36, c = 81$

$a^2 + b^2 + c^2 = 16 + 1296 + 6561 = 7873$

61. 3

Sol. Since maximum value of $\cos^{-1}x$ in $[0, 1] = \dfrac{\pi}{2}, \sum_{r=1}^{k}\cos^{-1}\beta_r = \dfrac{k\pi}{2}$ is possible if and only if each $\cos^{-1}\beta_r = \dfrac{\pi}{2} \Leftrightarrow \beta_r = 0$.

$A = \sum_{r=1}^{k}(\beta_r)^r = 0$

Thus, $\lim\limits_{x\to0} \dfrac{(1+x)^{1/3} - (1-2x)^{1/4}}{x + x^2}$

$= \lim\limits_{x\to0} \dfrac{\left(1 + \dfrac{x^2}{3} + 0\left(x^4\right)\right) - \left(1 - \dfrac{x}{2} + 0\left(x^2\right)\right)}{x(1+x)}$

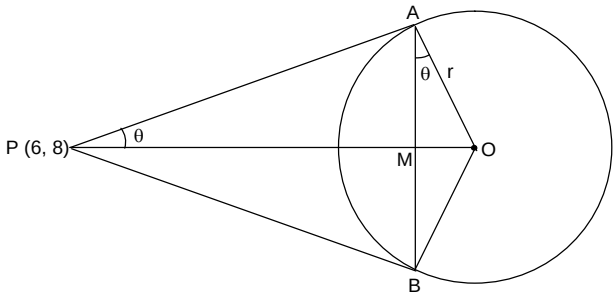
$= \lim\limits_{x\to0} \dfrac{\dfrac{1}{2} + \dfrac{x}{3} + 0\left(x^2\right)}{1+x} = \dfrac{1}{2}$. Thus the required limit is 249.

62. 5

Sol. Let O be the centre of the circle $x^2 + y^2 = r^2$ and P (6, 8) be the point outside the circle. Let θ be the angle between the tangent AP and OP. Clearly $\angle OAM = \theta$ so that $BM = r\cos\theta$ and $OM = r\sin\theta\left(0 < \theta < \dfrac{\pi}{2}\right)$.

Also $OP = \sqrt{36+64} = 10$ and $r = 10\sin\theta$.

If A is the area of the triangle PAB, then



$A = 2$ (area of ΔPBM)

$= 2\left(\left(\frac{1}{2}\right)PM \times BM\right) = (OP - OM)BM$

$= (10 - r\sin\theta)r\cos\theta$

$= \left(10 - 10\sin^2\theta\right)(10\sin\theta\cos\theta)$

$\left(\because r = 10\sin\theta\right)$

$= 100\cos^3\theta\sin\theta$

$= 50\sin 2\theta\cos^2\theta$

$= \frac{25}{2}\left[\sin 4\theta + 2\sin 2\theta\right]$

So, $\frac{dA}{d\theta} = 50\left(\cos 4\theta + \cos 2\theta\right)$

$= 100\cos 3\theta\cos\theta$

For maximum value of θ, $\frac{dA}{d\theta} = 0$

$\Rightarrow \quad \cos 4\theta + \cos 2\theta = 0$

$\Rightarrow \quad 2\cos 3\theta\cos\theta = 0$

$\Rightarrow \quad 3\theta = \frac{\pi}{4}\left(0 < \theta < \frac{\pi}{2}\right)$

$\Rightarrow \quad \theta = \frac{\pi}{6}$

$\frac{d^2A}{d\theta^2} = -100\left(2\sin 4\theta + \sin 2\theta\right)$

$\left.\frac{d^2A}{d\theta^2}\right|_{\theta=\pi/6} = -100\left(\sqrt{3} + \frac{\sqrt{3}}{2}\right) < 0$

63. 4

Sol. (i) Putting $e^x = t$ in the given integral and taking e^x common in the integrand, we can write the given integral as $\int \frac{2t^4 + t^3 - 4t^2 + 4t + 2}{\left(t^2 + 4\right)\left(t^2 - 1\right)^2}dt$

$= \int \frac{2t^4 - 4t^2 + 2}{\left(t^2 + 4\right)\left(t^2 - 1\right)^2}dt + \int \frac{t^3 + 4t}{\left(t^2 + 4\right)\left(t^2 - 1\right)^2}dt$

$= 2\int \frac{\left(t^2 - 1\right)^2}{\left(t^2 + 4\right)\left(t^2 - 1\right)^2}dt + \int \frac{t\left(t^2 + 4\right)}{\left(t^2 + 4\right)\left(t^2 - 1\right)^2}dt$

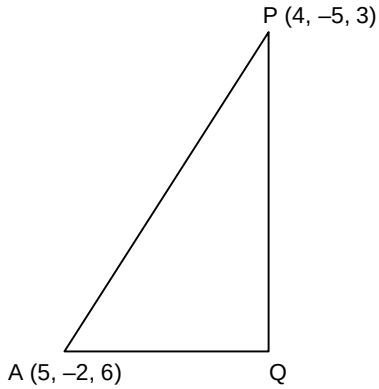
$= 2\int \frac{dt}{t^2 + 4} + \frac{1}{2}\int \frac{2t}{\left(t^2 - 1\right)^2}dt$

$$= \tan^{-1}\frac{t}{2} - \frac{1}{2\left(t^2-1\right)} + C$$

$$= \tan^{-1}\left(\frac{e^x}{2}\right) - \frac{1}{2\left(e^{2x}-1\right)} + C$$

64. 7

Sol. The given lines passes through A (5, –2, 6)



$$(PQ)^2 = (AP)^2 - (AQ)^2$$

$$(AP)^2 = (4-5)^2 + (-5+2)^2 + (3-6)^2 = 19$$

AQ is the projection of AP on the given line

$$\therefore AQ = (5-4)\frac{3}{\sqrt{50}} + (-2+5)\frac{(-4)}{\sqrt{50}} + (6-3)\left(\frac{-5}{\sqrt{50}}\right) = \frac{6}{\sqrt{50}}$$

$$\Rightarrow (PQ)^2 = 19 - \frac{36}{50} = \frac{457}{25}$$

$$\Rightarrow 25(PQ)^2 - 450 = 7$$

65. 3

Sol. Taking logarithm of both sides, we get $\log y = x\left[\log\left(1+\frac{1}{x}\right)\right]$

$$\Rightarrow \frac{1}{y}y_1(x) = \frac{x^2}{x+1}\left(-\frac{1}{x^2}\right) + \log\left(1+\frac{1}{x}\right)$$

$$= -\frac{1}{x+1} + \log\left(1+\frac{1}{x}\right) \quad \text{(i)}$$

Since $y(2) = \left(1+\frac{1}{2}\right)^2 = \frac{9}{4}$ so

$$y_1(2) = \left(\frac{9}{4}\right)\left(-\frac{1}{3} + \log\frac{3}{2}\right)$$

Multiplying (i) by y and then differentiating, we get

$$y_2(x) = y_1(x)\left(-\frac{1}{x+1} + \log\left(1+\frac{1}{x}\right)\right) + y(x)\times\left(\frac{1}{(x+1)^2} + \frac{x}{x+1}\left(-\frac{1}{x^2}\right)\right)$$

So $y_2(2) = y_1(2)\left(-\frac{1}{3} + \log\frac{3}{2}\right) + y(2)\left(\frac{1}{9} - \frac{1}{6}\right)$

$$= \left(\frac{9}{4}\right)\left(-\frac{1}{3} + \log\frac{3}{2}\right)^2 - \frac{1}{8}.$$

66. 1

Sol. Let $y = \frac{e^x - e^{-x}}{2} \Rightarrow e^{2x} - 1 = 2ye^x$

Therefore, $t^2 - 2yt - 1 = 0,\ t = e^x$

$\Rightarrow t = \dfrac{2y \pm \sqrt{4y^2+4}}{2} = y \pm \sqrt{y^2+1}$

$\Rightarrow x = \log\left(y + \sqrt{y^2+1}\right)$ (since $e^x > 0$)

$\therefore\ f^{-1}(x) = g(x) = \log\left(x + \sqrt{x^2+1}\right)$

$g\left(\dfrac{e^{1002}-1}{2e^{501}}\right) = \log\left(\dfrac{e^{1002}-1}{2e^{501}} + \dfrac{e^{1002}+1}{2e^{501}}\right)$

$= \log e^{501} = 501$

SECTION – D

67. 24850.00

Sol. $\left(1-2hx+h^2\right)^{-1/2} = \left(1-h(2x-h)\right)^{-1/2}$

$= 1 + \dfrac{h}{2}(2x-h) + \dfrac{\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right)}{2}h^2(2x-h)^2 + \dfrac{1.3.5}{2.4.6}h^3(2x-h)^3 + ...$

$= 1 + xh + \dfrac{1}{2}\left(3x^2-1\right)h^2 + \dfrac{1}{2}\left(5x^3-3x\right)h^3 + ...$

$P_3(x) =$ Coefficient of h^3 in $\left(1-2hx+h^2\right)^{-1/2}$

$= \dfrac{1}{2}\left(5x^3-3x\right).$

$P_3(10) = \dfrac{1}{2}(5000-30) = 2485$

68. 26916.72

Sol. Since $x(1-x) = \dfrac{1}{4} - \left(x - \dfrac{1}{2}\right)^2$, so put $x = \dfrac{1}{2} + \left(\dfrac{1}{2}\right)\sin\theta$.

Therefore, $\displaystyle\int_0^1 \frac{dx}{(1+x)(2+x)\sqrt{x(1-x)}}$

$= \displaystyle\int_{-\pi/2}^{\pi/2} \frac{\left(\frac{1}{2}\right)\cos\theta\, d\theta}{\left(\frac{1}{4}(1+\sin\theta)^2 + \frac{3}{2}(1+\sin\theta) + 2\right)\frac{1}{2}\cos\theta}$

$= 4\displaystyle\int_0^{\pi} \frac{d\phi}{\cos^2\phi - 8\cos\phi + 15}$ where, $\phi = \theta + \dfrac{\pi}{2}$

$= \dfrac{4}{2}\left[\displaystyle\int_0^{\pi} \frac{d\phi}{\cos\phi - (4+1)} - \int_0^{\pi} \frac{d\phi}{\cos\phi - (4-1)}\right]$

Let $I_1 = \int_0^{\pi} \frac{d\phi}{\cos\phi - a}$ where $a > 1$

Put $\tan\left(\frac{\phi}{2}\right) = t$, $d\phi = 2\frac{dt}{\left(1+t^2\right)}$.

$$\therefore I_1 = \int_0^{\infty} \frac{\frac{2dt}{\left(1+t^2\right)}}{\frac{\left(1-t^2\right)}{\left(1+t^2\right)} - a} = \frac{-2}{1+a}\int_0^{\infty} \frac{dt}{r^2 + \frac{(a-1)}{(a+1)}}$$

$$= \frac{-2}{1+a}\sqrt{\frac{a+1}{a-1}}\left(\tan^{-1} t\sqrt{\frac{a+1}{a-1}}\right)\Bigg]_0^{\infty}$$

$$= \frac{-2}{\sqrt{a^2-1}}.\frac{\pi}{2} = \frac{-\pi}{\sqrt{a^2-1}}$$

Thus $I = 2\left[-\frac{\pi}{\sqrt{24}} + \frac{\pi}{\sqrt{8}}\right]$

$$= \pi\left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{6}}\right) = \pi\frac{\left(\sqrt{3}-1\right)}{\sqrt{6}}$$

69. 24869.83

Sol. Let AB be the line whose equation is $\frac{x}{97} + \frac{y}{79} = 1$ where A is (97, 0) and B is (0, 79). Equation of the other diagonal passing through $\left(\frac{97}{2}, \frac{79}{2}\right)$, the mid point of AB and perpendicular to AB is $y - \frac{79}{2} = \frac{97}{79}\left(x - \frac{97}{2}\right)$ (1)

Let the coordinates of the vertex C of the other diagonal be (x_1, y_1) then (x_1, y_1) lies on (1) and $BC^2 = \frac{AB^2}{2}$

$$\Rightarrow 2\left[x_1^2 + (y_1 - 79)^2\right] = (97)^2 + (79)^2$$

$$\Rightarrow 2\left[\left(1 + \frac{97^2}{79^2}\right)x_1^2 - 2\times\frac{97}{79}\left(\frac{97^2+79^2}{2\times 79}\right)x_1 + \left(\frac{(97)^2+(79)^2}{2(79)}\right)^2\right]$$

$$\Rightarrow 4x_1^2 - 4\times 97x_1 + (97)^2 + (79)^2 = 2\times(79)^2$$

$$\Rightarrow (2x_1 - 97)^2 = (79)^2$$

$$\Rightarrow 2x_1 = 97 \pm 79 \Rightarrow x_1 = \frac{97\pm 79}{2} y_1 = \frac{97\pm 79}{2}$$

$\therefore$ The coordinates of the vertices of the other diagonal are (88, 88) and (9, 9)

$\Rightarrow p_1 = 88, p_2 = 9$

$\Rightarrow p_1 + p_2 = 97$